

1. Variable Classification

CONTINUOUS (NUMERICAL) FEATURES:

- bmi
- age_at_nephrectomy
- hospitalization
- estimated_blood_loss
- operative_time
- radiographic_size
- pathologic_size
- last_preop_egfr.days_before_nephrectomy
- last_preop_egfr.value
- first_postop_egfr.days_after_nephrectomy
- first_postop_egfr.value
- last_postop_egfr.days_after_nephrectomy
- last_postop_egfr.value

DISCRETE (CATEGORICAL) FEATURES:

- gender
- er_visit
- readmission
- surgery_type
- surgical_approach
- surgical_procedure
- clavian_surgical_complications
- smoking_history
- chewing_tobacco_use
- alcohol_use
- cytoreductive
- malignant
- pathology_t_stage
- pathology_n_stage
- pathology_m_stage
- tumor_histologic_subtype
- tumor_necrosis
- tumor_isup_grade
- positive_resection_margins
- sarcomatoid_features
- rhabdoid_features
- aua_risk_score
- comorbidities.myocardial_infarction
- comorbidities.congestive_heart_failure
- comorbidities.peripheral_vascular_disease
- comorbidities.cerebrovascular_disease
- comorbidities.dementia
- comorbidities.copd
- comorbidities.connective_tissue_disease
- comorbidities.peptic_ulcer_disease
- comorbidities.uncomplicated_diabetes_mellitus
- comorbidities.diabetes_mellitus_with_end_organ_damage
- comorbidities.chronic_kidney_disease
- comorbidities.hemiplegia_from_stroke
- comorbidities.leukemia
- comorbidities.malignant_lymphoma
- comorbidities.localized_solid_tumor
- comorbidities.metastatic_solid_tumor
- comorbidities.mild_liver_disease
- comorbidities.moderate_to_severe_liver_disease
- comorbidities.aids
- intraoperative_complications.blood_transfusion
- intraoperative_complications.injury_to_surrounding_organ

2. Data Trimming & Feature Selection

COLUMNS DROPPED DUE TO EXCESSIVE MISSINGNESS (>40%)

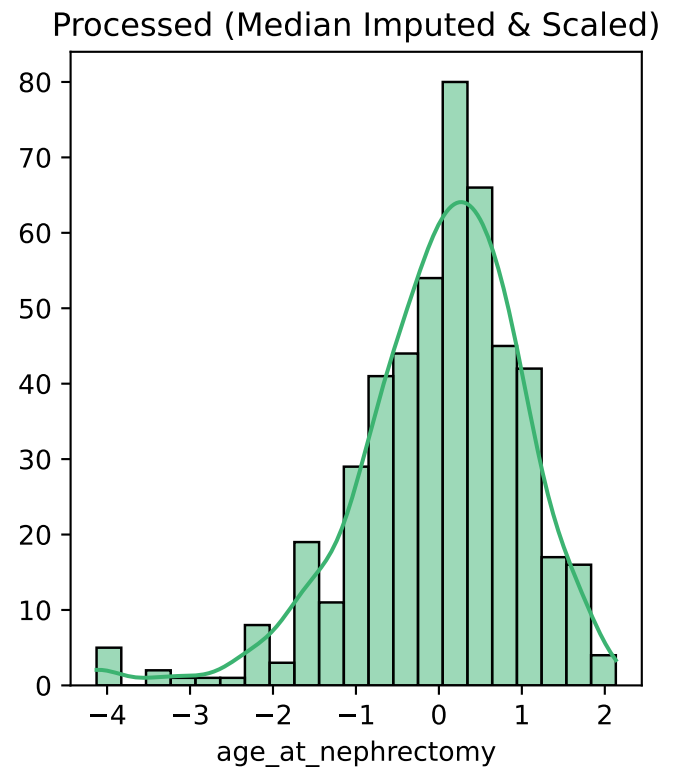
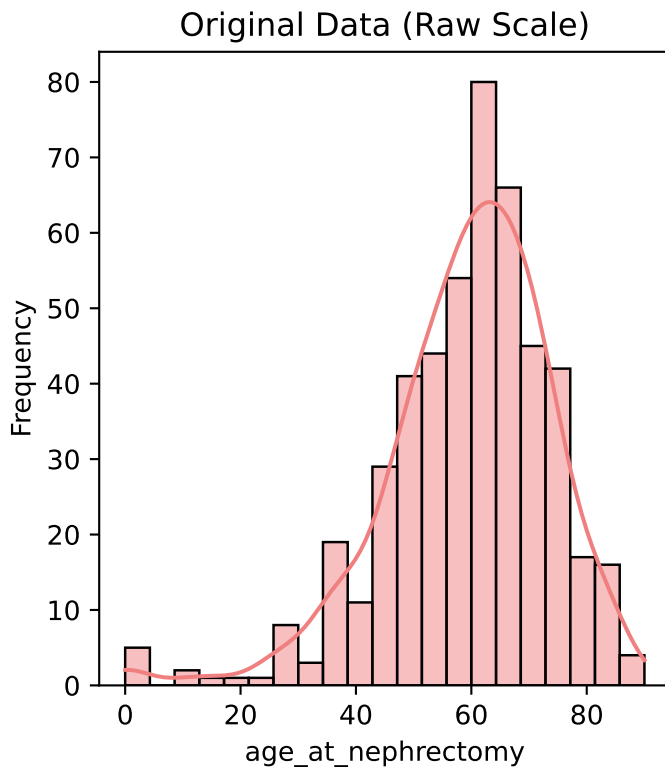
- ischemia_time : 47.03% Missing
- age_when_quit_smoking : 69.33% Missing
- pack_years : 48.26% Missing

MACHINE LEARNING JUSTIFICATION:

In Survival Analysis and Neural Networks, excessive missingness introduces critical flaws:

1. Noise Injection: When a column is >40% missing, imputation (guessing the value) dominates the actual signal.
2. Spurious Correlations: The model may learn the 'imputation rule' rather than finding a real physiological link to patient survival.
3. Multi-Modal Balance: Because you are fusing this tabular data with highly dense OmniRad image features, sparse/noisy tabular columns can destabilize the MLP gradients. Dropping them forces the network to rely on clean data.

Continuous Feature: age_at_nephrectomy



STATISTICAL SUMMARY:

Original Data (Before Processing):

- Missing Values : 0
- Mean : 59.33
- Std Dev : 14.39
- Min : 0.00
- Max : 90.00

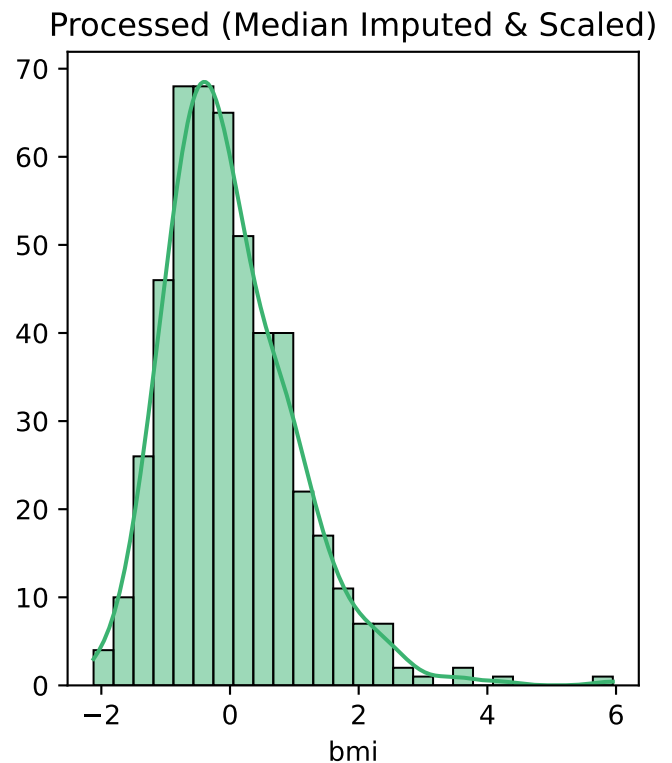
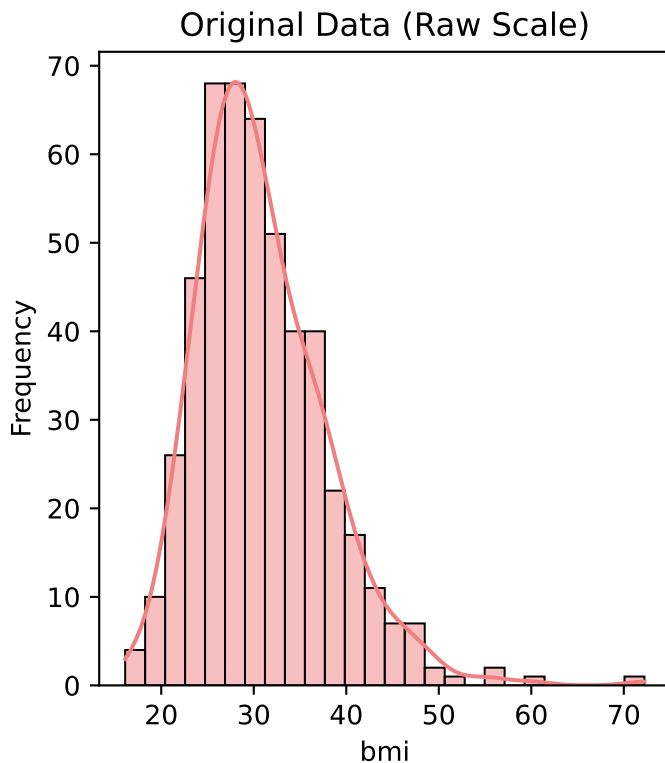
Processed Data (After Standardization):

- Missing Values : 0
- Mean : -0.000000 (Approaching 0)
- Std Dev : 1.001024 (Approaching 1)
- Min : -4.13
- Max : 2.13

TRANSFORMATION PIPELINE:

1. Median Imputation: Replaced missing values with the median to resist outliers.
2. Standard Scaling : $Z = (x - \text{mean}) / \text{std.}$
This ensures the neural network treats this feature's scale equally relative to other features.

Continuous Feature: bmi



STATISTICAL SUMMARY:

Original Data (Before Processing):

- Missing Values : 1
- Mean : 30.85
- Std Dev : 6.97
- Min : 16.10
- Max : 72.23

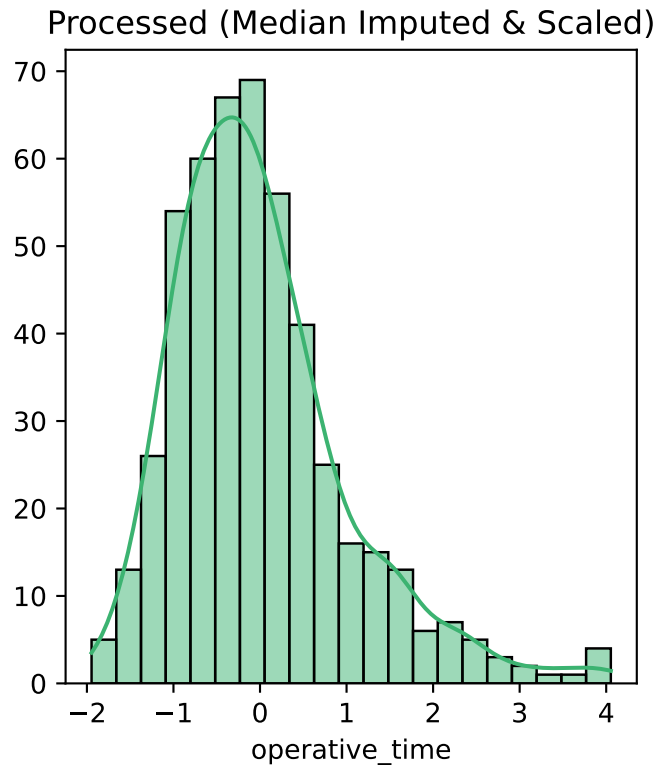
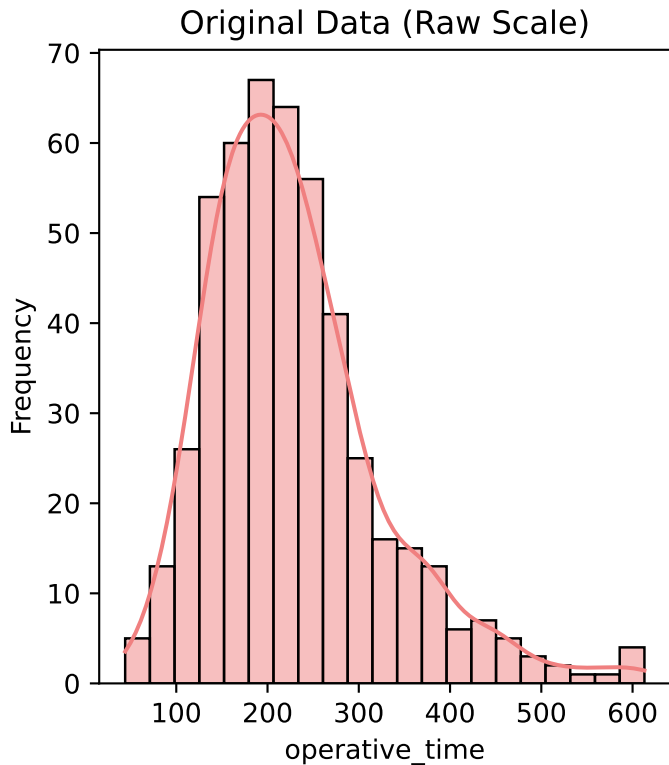
Processed Data (After Standardization):

- Missing Values : 0
- Mean : 0.000000 (Approaching 0)
- Std Dev : 1.001024 (Approaching 1)
- Min : -2.12
- Max : 5.95

TRANSFORMATION PIPELINE:

1. Median Imputation: Replaced missing values with the median to resist outliers.
2. Standard Scaling : $Z = (x - \text{mean}) / \text{std.}$
This ensures the neural network treats this feature's scale equally relative to other features.

Continuous Feature: operative_time



STATISTICAL SUMMARY:

Original Data (Before Processing):

- Missing Values : 5
- Mean : 228.73
- Std Dev : 95.41
- Min : 44.00
- Max : 613.00

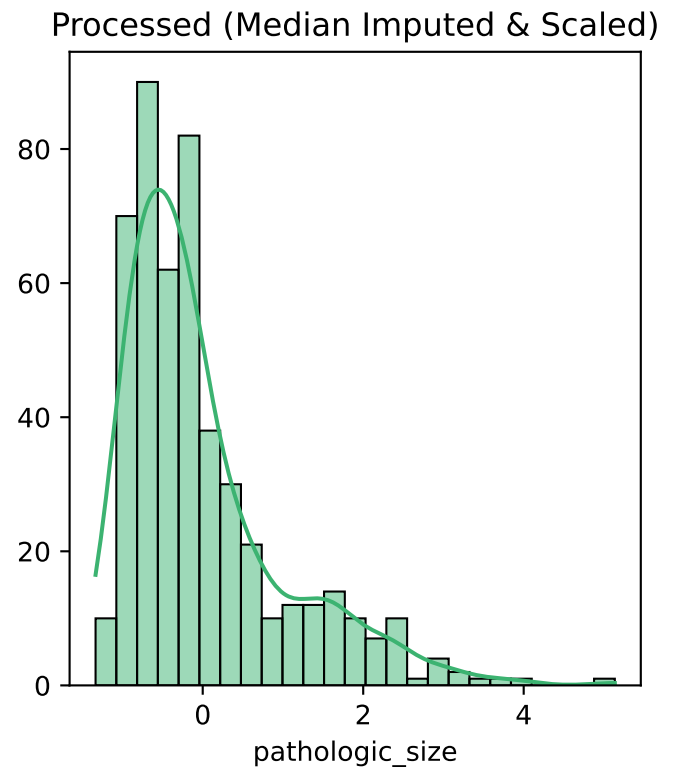
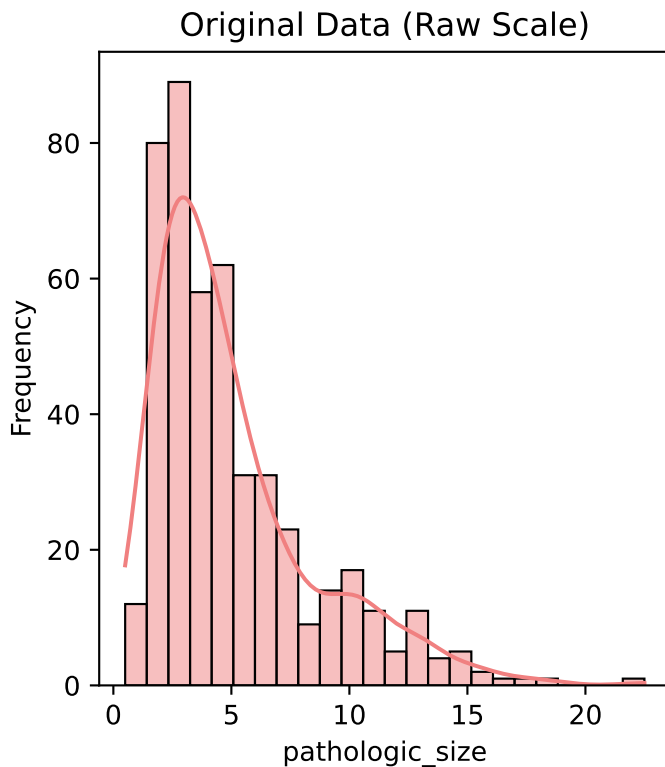
Processed Data (After Standardization):

- Missing Values : 0
- Mean : 0.000000 (Approaching 0)
- Std Dev : 1.001024 (Approaching 1)
- Min : -1.95
- Max : 4.05

TRANSFORMATION PIPELINE:

1. Median Imputation: Replaced missing values with the median to resist outliers.
2. Standard Scaling : $Z = (x - \text{mean}) / \text{std.}$
This ensures the neural network treats this feature's scale equally relative to other features.

Continuous Feature: pathologic_size



STATISTICAL SUMMARY:

Original Data (Before Processing):

- Missing Values : 21
- Mean : 5.08
- Std Dev : 3.47
- Min : 0.50
- Max : 22.50

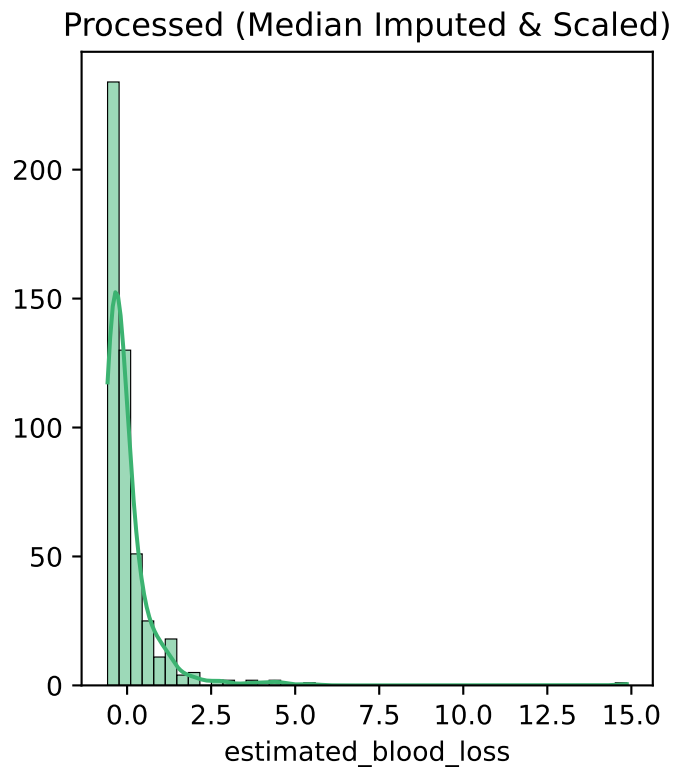
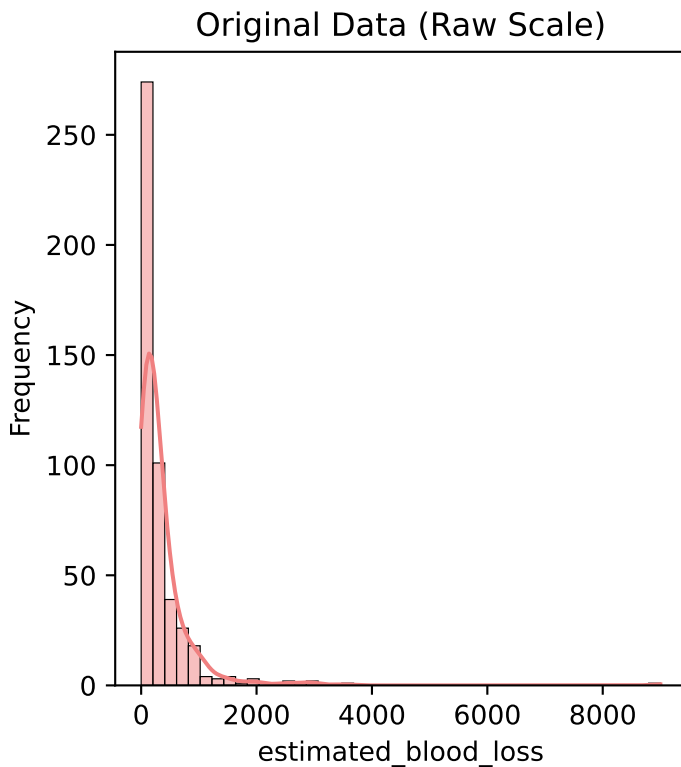
Processed Data (After Standardization):

- Missing Values : 0
- Mean : 0.000000 (Approaching 0)
- Std Dev : 1.001024 (Approaching 1)
- Min : -1.34
- Max : 5.14

TRANSFORMATION PIPELINE:

1. Median Imputation: Replaced missing values with the median to resist outliers.
2. Standard Scaling : $Z = (x - \text{mean}) / \text{std}$.
This ensures the neural network treats this feature's scale equally relative to other features.

Continuous Feature: estimated_blood_loss



STATISTICAL SUMMARY:

Original Data (Before Processing):

- Missing Values : 9
- Mean : 341.02
- Std Dev : 588.41
- Min : 0.00
- Max : 9000.00

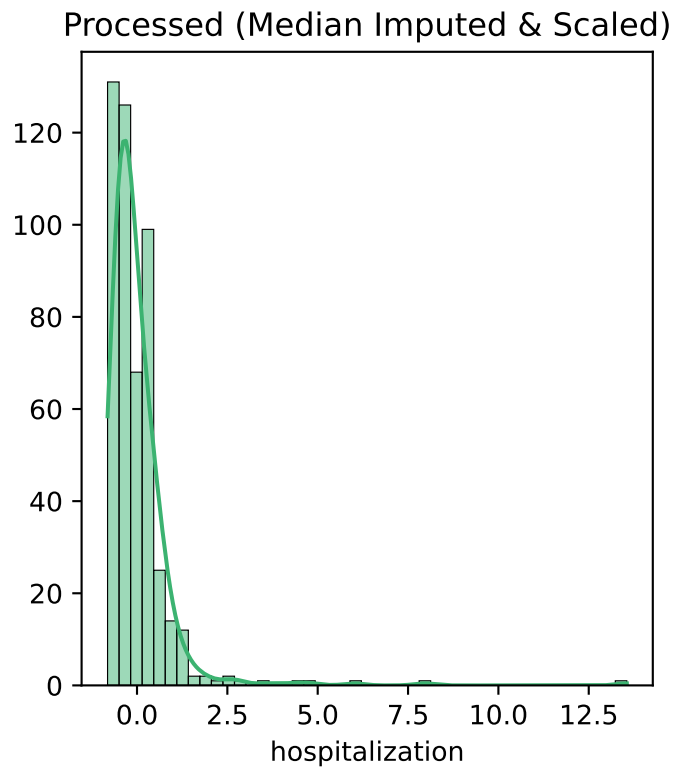
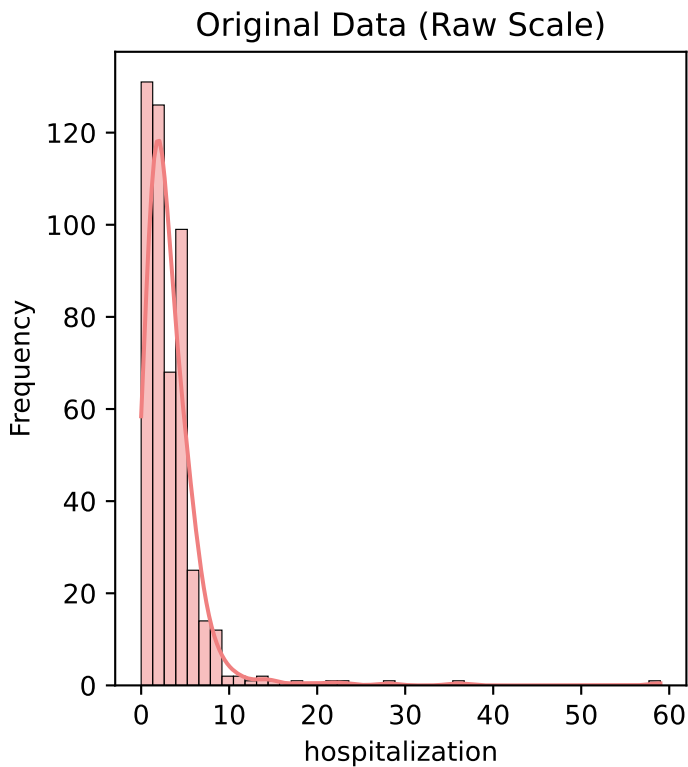
Processed Data (After Standardization):

- Missing Values : 0
- Mean : -0.000000 (Approaching 0)
- Std Dev : 1.001024 (Approaching 1)
- Min : -0.58
- Max : 14.87

TRANSFORMATION PIPELINE:

1. Median Imputation: Replaced missing values with the median to resist outliers.
2. Standard Scaling : $Z = (x - \text{mean}) / \text{std.}$
This ensures the neural network treats this feature's scale equally relative to other features.

Continuous Feature: hospitalization



STATISTICAL SUMMARY:

Original Data (Before Processing):

- Missing Values : 0
- Mean : 3.35
- Std Dev : 4.11
- Min : 0.00
- Max : 59.00

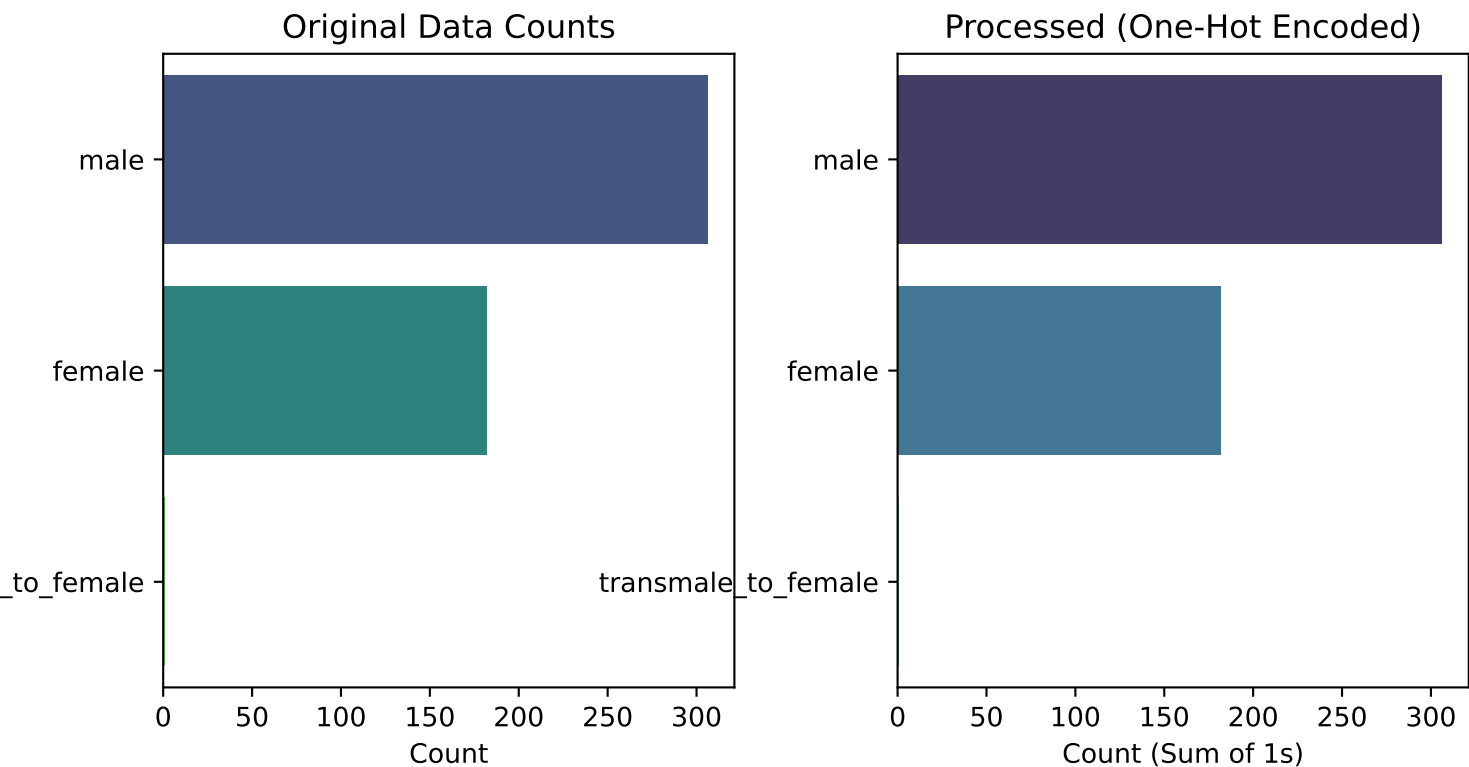
Processed Data (After Standardization):

- Missing Values : 0
- Mean : -0.000000 (Approaching 0)
- Std Dev : 1.001024 (Approaching 1)
- Min : -0.81
- Max : 13.55

TRANSFORMATION PIPELINE:

1. Median Imputation: Replaced missing values with the median to resist outliers.
2. Standard Scaling : $Z = (x - \text{mean}) / \text{std.}$
This ensures the neural network treats this feature's scale equally relative to other features.

Discrete Feature: gender



STATISTICAL SUMMARY & TRANSFORMATION:

Original Data Missing Values: 0

One-Hot Encoding Expansion:

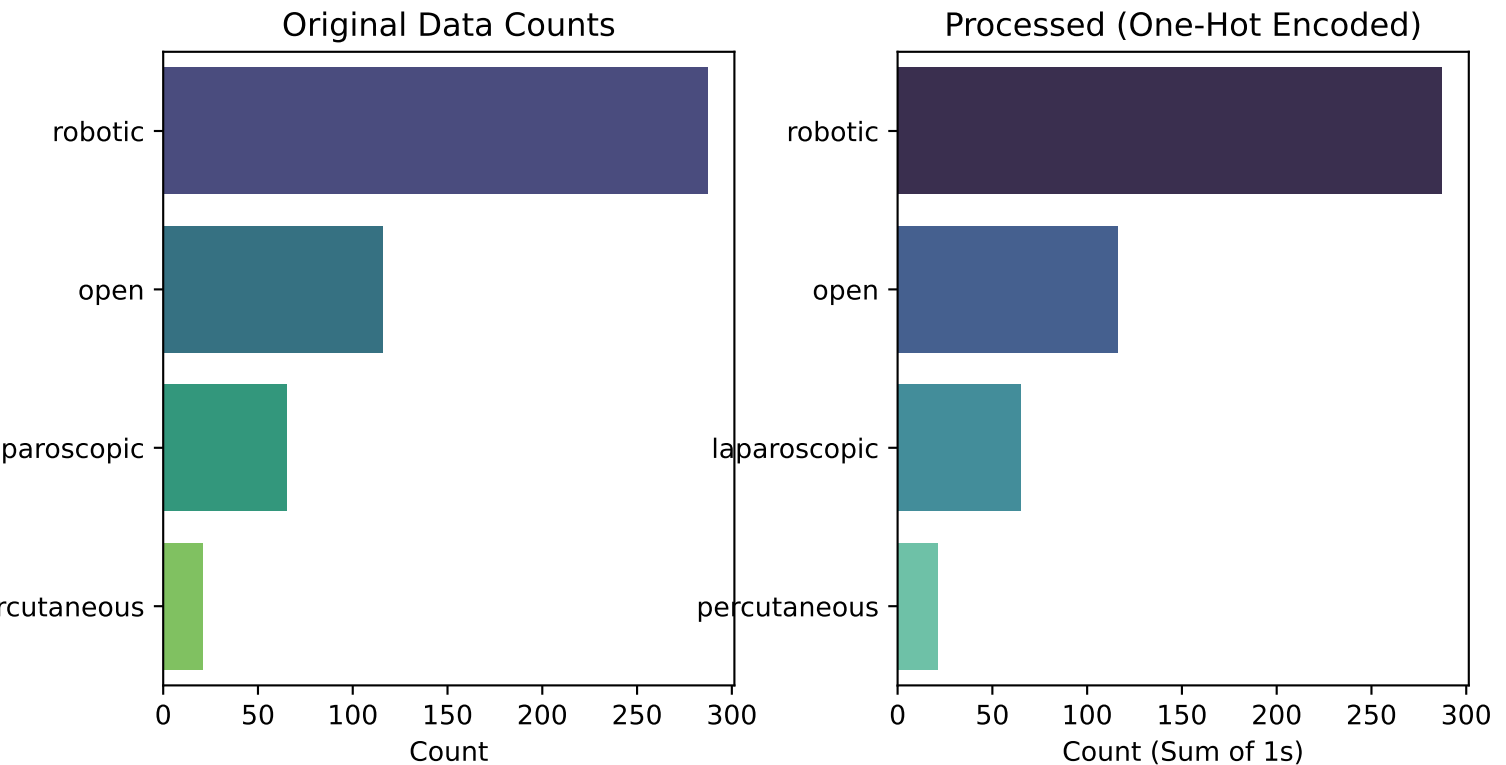
The original column 'gender' was expanded into 3 binary vectors.

- gender_male : 306 instances (1s)
- gender_female : 182 instances (1s)
- gender_transmale_to_female : 1 instances (1s)

TRANSFORMATION PIPELINE:

1. Constant Imputation: NaN values were explicitly replaced with 'Missing'.
2. One-Hot Encoding: Converted string labels into orthogonal binary vectors. This allows the ML model to assign distinct weights to each category without assuming any mathematical order (e.g., Male is not 'greater' than Female).

Discrete Feature: surgery_type



STATISTICAL SUMMARY & TRANSFORMATION:

Original Data Missing Values: 0

One-Hot Encoding Expansion:

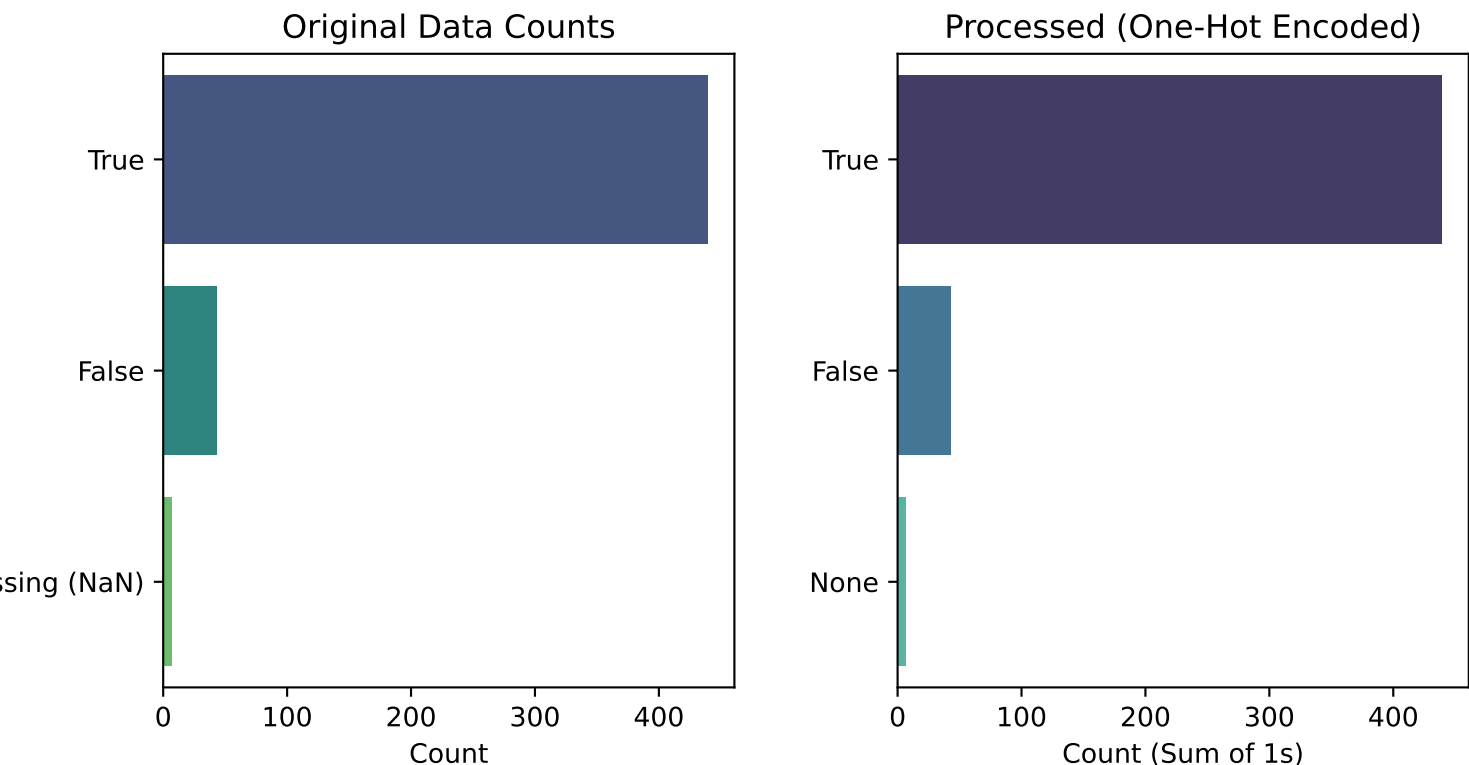
The original column 'surgery_type' was expanded into 4 binary vectors.

- surgery_type_robotic : 287 instances (1s)
- surgery_type_open : 116 instances (1s)
- surgery_type_laparoscopic : 65 instances (1s)
- surgery_type_percutaneous : 21 instances (1s)

TRANSFORMATION PIPELINE:

1. Constant Imputation: NaN values were explicitly replaced with 'Missing'.
2. One-Hot Encoding: Converted string labels into orthogonal binary vectors. This allows the ML model to assign distinct weights to each category without assuming any mathematical order (e.g., Male is not 'greater' than Female).

Discrete Feature: malignant



STATISTICAL SUMMARY & TRANSFORMATION:

Original Data Missing Values: 7

One-Hot Encoding Expansion:

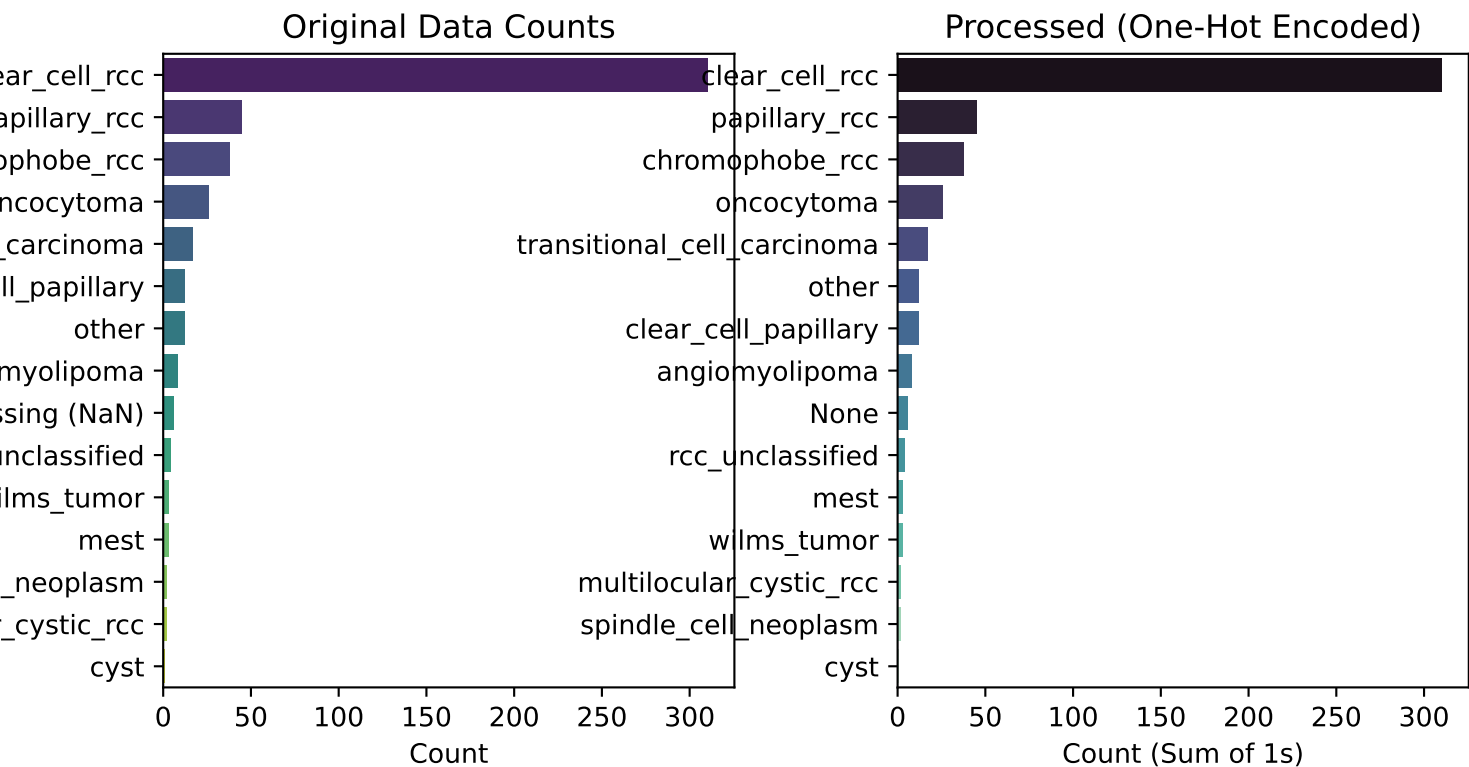
The original column 'malignant' was expanded into 3 binary vectors.

- malignant_True : 439 instances (1s)
- malignant_False : 43 instances (1s)
- malignant_None : 7 instances (1s)

TRANSFORMATION PIPELINE:

1. Constant Imputation: NaN values were explicitly replaced with 'Missing'.
2. One-Hot Encoding: Converted string labels into orthogonal binary vectors. This allows the ML model to assign distinct weights to each category without assuming any mathematical order (e.g., Male is not 'greater' than Female).

Discrete Feature: tumor_histologic_subtype



STATISTICAL SUMMARY & TRANSFORMATION:

Original Data Missing Values: 6

One-Hot Encoding Expansion:

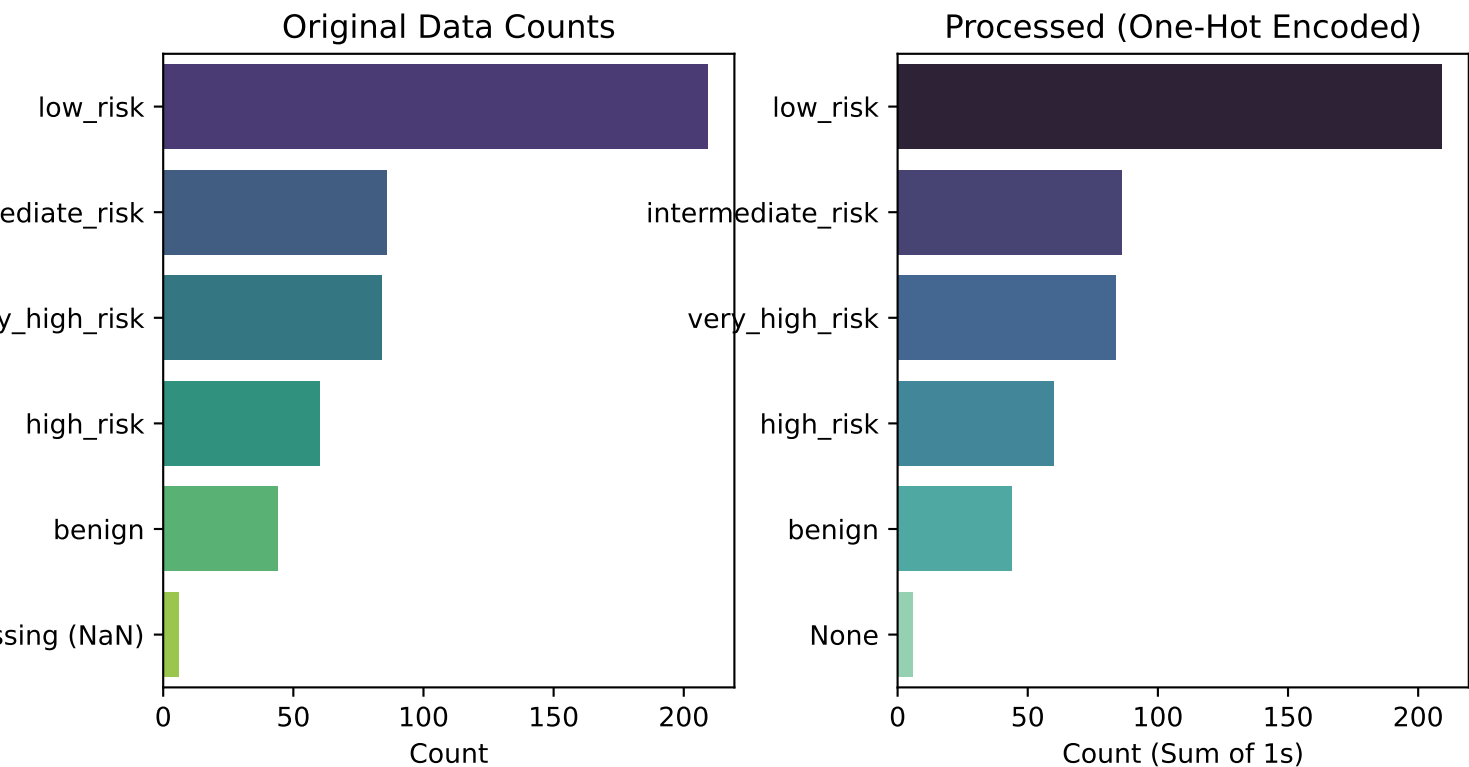
The original column 'tumor_histologic_subtype' was expanded into 15 binary vectors.

- tumor_histologic_subtype_clear_cell_rcc : 310 instances (1s)
- tumor_histologic_subtype_papillary_rcc : 45 instances (1s)
- tumor_histologic_subtype_chromophobe_rcc : 38 instances (1s)
- tumor_histologic_subtype_oncocyoma : 26 instances (1s)
- tumor_histologic_subtype_transitional_cell_carcinoma : 17 instances (1s)
- tumor_histologic_subtype_other : 12 instances (1s)
- tumor_histologic_subtype_clear_cell_papillary : 12 instances (1s)
- tumor_histologic_subtype_angiomyolipoma : 8 instances (1s)
- tumor_histologic_subtype_None : 6 instances (1s)
- tumor_histologic_subtype_rcc_unclassified : 4 instances (1s)
- tumor_histologic_subtype_mest : 3 instances (1s)
- tumor_histologic_subtype_wilms_tumor : 3 instances (1s)
- tumor_histologic_subtype_multilocular_cystic_rcc : 2 instances (1s)
- tumor_histologic_subtype_spindle_cell_neoplasm : 2 instances (1s)
- tumor_histologic_subtype_cyst : 1 instances (1s)

TRANSFORMATION PIPELINE:

1. Constant Imputation: NaN values were explicitly replaced with 'Missing'.
2. One-Hot Encoding: Converted string labels into orthogonal binary vectors. This allows the ML model to assign distinct weights to each category without assuming any mathematical order (e.g., Male is not 'greater' than Female).

Discrete Feature: aua_risk_score



STATISTICAL SUMMARY & TRANSFORMATION:

Original Data Missing Values: 6

One-Hot Encoding Expansion:

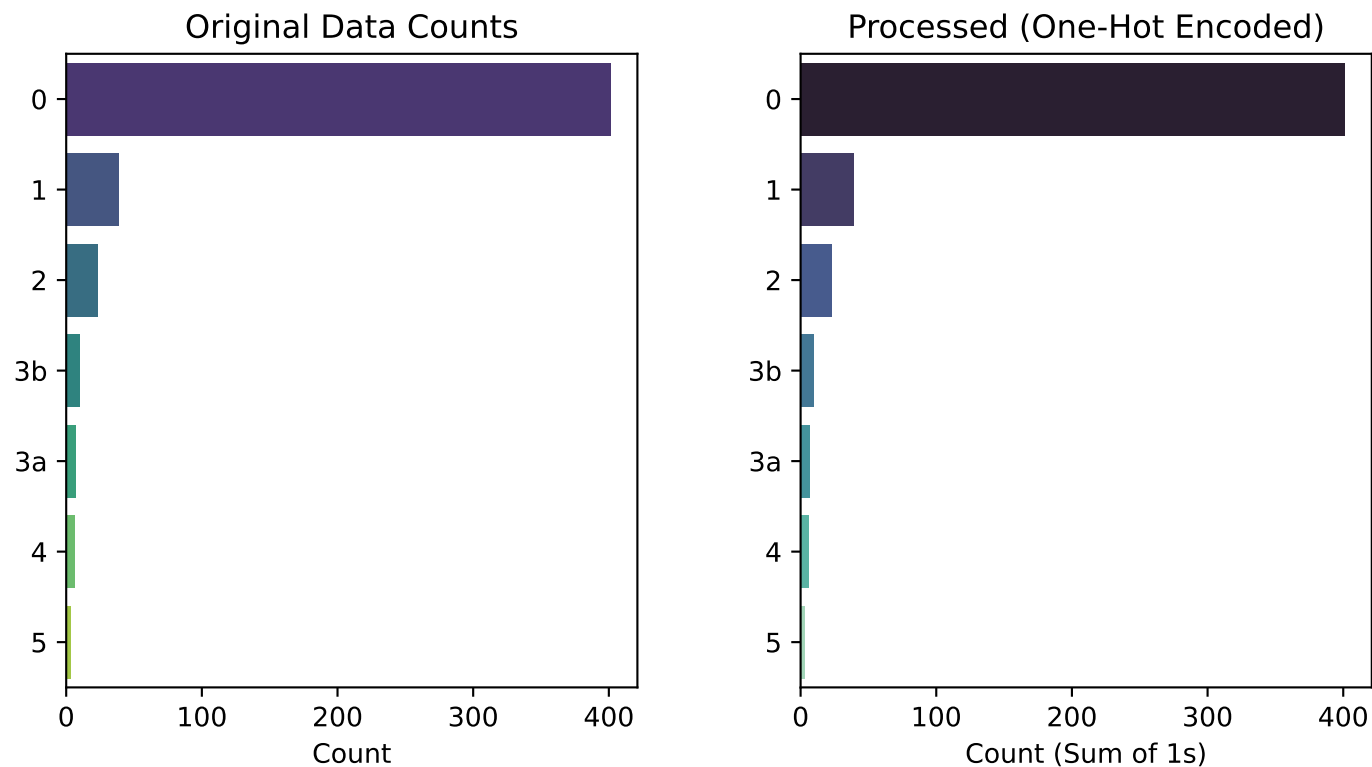
The original column 'aua_risk_score' was expanded into 6 binary vectors.

- aua_risk_score_low_risk : 209 instances (1s)
- aua_risk_score_intermediate_risk : 86 instances (1s)
- aua_risk_score_very_high_risk : 84 instances (1s)
- aua_risk_score_high_risk : 60 instances (1s)
- aua_risk_score_benign : 44 instances (1s)
- aua_risk_score_None : 6 instances (1s)

TRANSFORMATION PIPELINE:

1. Constant Imputation: NaN values were explicitly replaced with 'Missing'.
2. One-Hot Encoding: Converted string labels into orthogonal binary vectors. This allows the ML model to assign distinct weights to each category without assuming any mathematical order (e.g., Male is not 'greater' than Female).

Discrete Feature: clavien_surgical_complications



STATISTICAL SUMMARY & TRANSFORMATION:

Original Data Missing Values: 0

One-Hot Encoding Expansion:

The original column 'clavien_surgical_complications' was expanded into 7 binary vectors.

- clavien_surgical_complications_0 : 401 instances (1s)
- clavien_surgical_complications_1 : 39 instances (1s)
- clavien_surgical_complications_2 : 23 instances (1s)
- clavien_surgical_complications_3b : 10 instances (1s)
- clavien_surgical_complications_3a : 7 instances (1s)
- clavien_surgical_complications_4 : 6 instances (1s)
- clavien_surgical_complications_5 : 3 instances (1s)

TRANSFORMATION PIPELINE:

1. Constant Imputation: NaN values were explicitly replaced with 'Missing'.
2. One-Hot Encoding: Converted string labels into orthogonal binary vectors. This allows the ML model to assign distinct weights to each category without assuming any mathematical order (e.g., Male is not 'greater' than Female).